

Hassaan Mustafavi

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ABOUT ME

Applied AI / Automation Engineer with **2 years** of experience building AI agents, applications, and business automation systems using both Python-based development and low/no-code platforms. Experienced in backend integrations, workflow automation, debugging, and custom development, with additional work across SaaS, machine learning and forecasting, web automation, and data workflows. **Top Rated on Upwork with 100% JSS**, reflecting consistent delivery across real-world projects.

SKILLS & TOOLS

Applied AI & Agent Systems: LLM Applications, RAG, Agentic & Multimodal Systems, LangGraph, OpenAI Agents SDK, CrewAI, MCP, OpenAI, Claude, Vapi, Retell AI

Automation, Backend & Data: Python, FastAPI, REST APIs, n8n, Make, Zapier, CRM/SaaS Integrations, PostgreSQL, Supabase, pgvector/Qdrant, OCR/Document Processing, Data Pipelines, Webhooks & Async Processing

ML, Web/System Automation & Deployment: PyTorch, OpenCV, YOLOv8, Selenium, Playwright, Scrapy, PyAutoGUI, Docker, AWS EC2/S3/Lambda, Cloud Deployment

EXPERIENCE

Independent	Remote
AI Automation Engineer	Dec 2024 – Present
<i>Built and deployed AI/Automation systems for 16+ SMBs across automotive, real estate, health tech, SaaS, hospitality, and service industries.</i>	
TCL Pakistan	Lahore, Pakistan
AI/ML Engineer - Contract	Mar 2025 – Jul 2025
<i>Developed MVPs, an SOP Monitoring System that streams live station feeds and tracks technician compliance, and a Forecasting Tool that predicts store unit demand using KPIs, weather data, and dynamic business conditions.</i>	

FEATURED PROJECTS

- AI-Driven Spare Parts Automation System:** Built an end-to-end n8n workflow for request classification, routing, part lookup, inventory search, pricing, and order processing, integrating EPC catalog (APIWIN17), VIN/VAN decoding, and real-time inventory with up to 65%-time savings and 93% part-number accuracy.
- RAG Chatbot for Insurance Brokerage:** Built a chatbot to answer internal team queries from policy PDFs, coverage docs, claim guidelines, pricing sheets, and FAQs, with source-grounded responses and human escalation using LangChain, OpenAI, Supabase Vector DB, OCR, and FAST API.
- AI Voice Agents:** Delivered multiple voice agents for SMBs, including a real estate agent that automated buyer/property intake, lead qualification, meeting booking, confirmations, reminders, human escalation, and Monday.com CRM sync using Vapi, n8n, OpenAI, Supabase, WABA and internal APIs.
- Claude-Powered Marketing Performance Agent:** Built an ads ops agent that pulls daily Meta/Google campaign stats, uses Claude API to summarize spend, CTR, CPC, CPA, ROAS, leads, and conversions, and triggers rule-based pause/alert workflows via n8n when budget, CPA, or ROAS thresholds are breached.

5. **Desktop App Automation:** Built a Windows automation app to generate PDF reports from legacy desktop software across 4 versions and 5 languages, automating Google Sheets input, form filling, dropdowns, validation, OCR-based clicks, and report export; deployed on Windows Server 2016 with crash recovery, auto-retry, safe bypass, and per-entry status tracking, reducing 30-minute manual workflows.
6. **Web Automation & Data Extraction System:** Built a cloud-deployed scraping workflow for Auction.com to log into multiple accounts, track foreclosure listings, capture bids, detect listing changes, and export leads to Google Sheets with scheduling, Chrome profiles, proxies, CAPTCHA handling, and Zoho/Gmail tracking.
7. **AI-Powered Patient Management System:** Built an appointment and communication system with a WhatsApp AI agent for scheduling, patient follow-ups, and automated visit summaries, increasing bookings by 15% and reducing follow-up questions by 40%.

ADDITIONAL PROJECTS

1. **AI Data Extraction System:** Built an internal SaaS tool to process large PDFs (125+ pages) and Excel reports, combining OCR and structured data pipelines to extract, clean, classify, and convert incoming quotes/requests into 6-layer decision-ready outputs.
2. **LinkedIn Lead Gen Agent:** Built an n8n workflow that finds LinkedIn prospects from role/company/location prompts, enriches emails with Hunter.io, uses Gemini to personalize outreach from public profile data, and saves Gmail drafts plus tracking rows in Google Sheets.
3. **Lead Gen Email Automation System:** Built a GPT-4o-powered email automation system for a custom webmail platform to detect unreplied customer messages and send one-time context-aware replies using Python, Selenium, prompt engineering, and Heroku deployment.
4. **Web Scraping:** Collected 115k+ unique property listings from Zameen.com and Graana.com, Harvested contacts from 13k+ libraries, and 25k+ synagogues, churches and other green team sites and more with AI assisted approach.
5. **GharBazar - AI-Powered Real Estate App:** Built a full-stack real estate Mobile app for buying, selling, and renting properties, with MCDM-based personalized search, AI price prediction, rent forecasting, and scraped property data pipelines using Flutter, Firebase, Python, FastAPI, PyTorch, and Prophet.
6. **SOP Monitoring System:** Built a real-time safety compliance system (MVP) using YOLOv8, OpenCV, and Python to detect technician presence and safety gear, trained on a custom 5k+ image dataset, with PyQt5 alerts, CSV/Pandas sync, async processing, and executable packaging.
7. **AI Demand Forecasting System:** Delivered a forecasting system for TCL (MVP) after-sales operations to predict stock needs, trigger low-stock alerts, and improve restocking decisions over time using operational KPIs, historical demand data, and reinforcement learning, achieving 73% forecast accuracy.
8. **Multimodal WhatsApp AI Agent:** Built a memory-enabled WhatsApp agent that handles text, voice, and image interactions in real time, with voice transcription/replies, image analysis/generation, short-term SQLite memory, and long-term Qdrant retrieval using LangGraph, Groq, Whisper, FLUX, Docker, and Qdrant.
9. **ERP Desktop Automation for Data Extraction:** Built a Python/PyAutoGUI automation app for an Arabic ERP UI to read part numbers from Excel, search billing/ERP screens, capture non-exportable scrolling table results, and export structured data into a single sheet.

EDUCATION

National University of Computer and Emerging Sciences (FAST)
Bachelors in Computer Science

Lahore, Pakistan
 2021 – 2025